

Homework (Habanero)

- Q1.** What is the primary purpose of a stem-and-leaf plot in healthcare statistics?
- (A) To compare treatment outcomes
(B) To display the distribution of patient data
(C) To calculate medication dosages
(D) To identify trends in cell growth
(E) To analyze medical imaging data
- Q2.** A researcher collects data on the 0 of hours patients spend in the hospital. The stem-and-leaf plot for this data shows stems of 2, 3, and 4. What can be inferred about the data?
- (A) Most patients spend less than 20 hours in the hospital
(B) Most patients spend between 20 and 30 hours in the hospital
(C) Most patients spend between 30 and 40 hours in the hospital
(D) Most patients spend more than 40 hours in the hospital
(E) The data is uniformly distributed
- Q3.** Which of the following is a benefit of using a stem-and-leaf plot to display data on of medication?
- (A) It allows for easy calculation of mean and median
(B) It provides a clear visual representation of the data distribution
(C) It enables identification of outliers and skewness
(D) It facilitates comparison with other datasets
(E) All of the above
- Q4.** A stem-and-leaf plot shows the following data: {3|1, 2, 4, 5, 7, 9}. What is the value of this dataset?
- (A) 31
(B) 32
(C) 34
(D) 35
(E) 37
- Q5.** Which type of data is best suited for a stem-and-leaf plot?
- (A) Categorical data
(B) Numerical data with a small range
(C) Numerical data with a large range
(D) Time-series data
(E) Geospatial data
- Q6.** What does the 'stem' in a stem-and-leaf plot typically represent?
- (A) The most significant digit of the data value
(B) The least significant digit of the data value
(C) The median of the data values
(D) The standard deviation of the data values
(E) The range of the data values
- Q7.** A researcher wants to compare the distribution of pressure values between two patient groups. Which of the following is a suitable method?
- (A) Create two separate stem-and-leaf plots
(B) Create a single stem-and-leaf plot with different colors for each group
(C) Calculate the mean and standard deviation for each group
(D) Use a histogram to display the data
(E) Use a box plot to display the data
- Q8.** What can be inferred about a dataset with a stem-and-leaf plot that shows a large gap between the stems?
- (A) The data is normally distributed
(B) The data is skewed to the left
(C) The data is skewed to the right
(D) The data has a bimodal distribution
(E) The data has outliers

Open Ended Questions (FRQ)

- Q9.** Create a stem-and-leaf plot for the following data on 0 of patients treated per day: 12, 15, 18, 20, 22, 25, 30, 32, 35, 38, 40, 42. What can be inferred about the distribution of this data? Provide at least 3 observations.
- Q10.** A stem-and-leaf plot shows the following data on of medication (): {2|0, 2, 5, 8}, {3|1, 4, 6, 9}. Calculate the , , and of this dataset. Provide your work and explain the significance of each value.

Chef's Tips

- Tip 1: Ensure students understand the purpose of stem-and-leaf plots in data analysis
- Tip 2: Encourage students to practice creating and interpreting stem-and-leaf plots with different types of data
- Tip 3: Remind students to consider the context and limitations of the data when making inferences from a stem-and-leaf plot